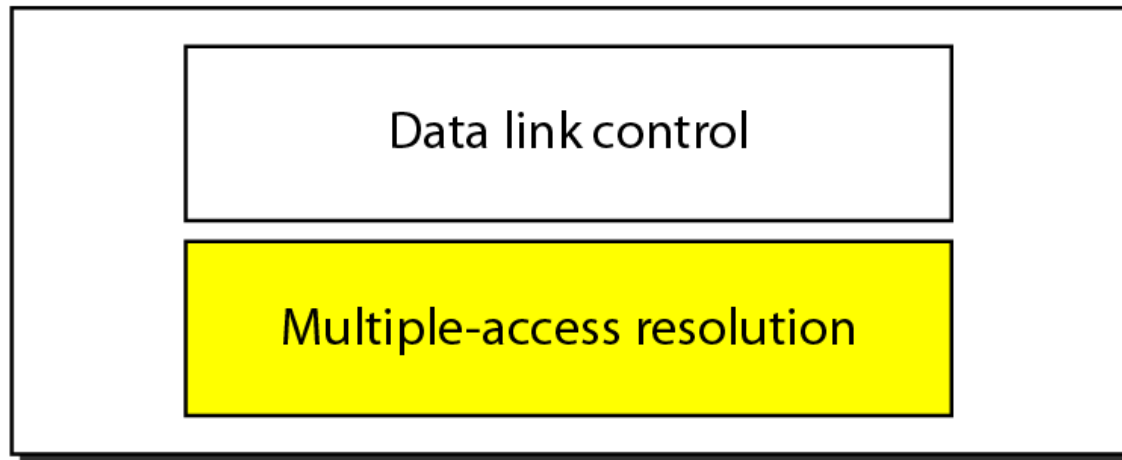


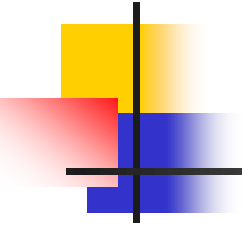
Wireless network

Multiple Radio Access

Data link layer divided into two functionality-oriented sublayers

Data link layer



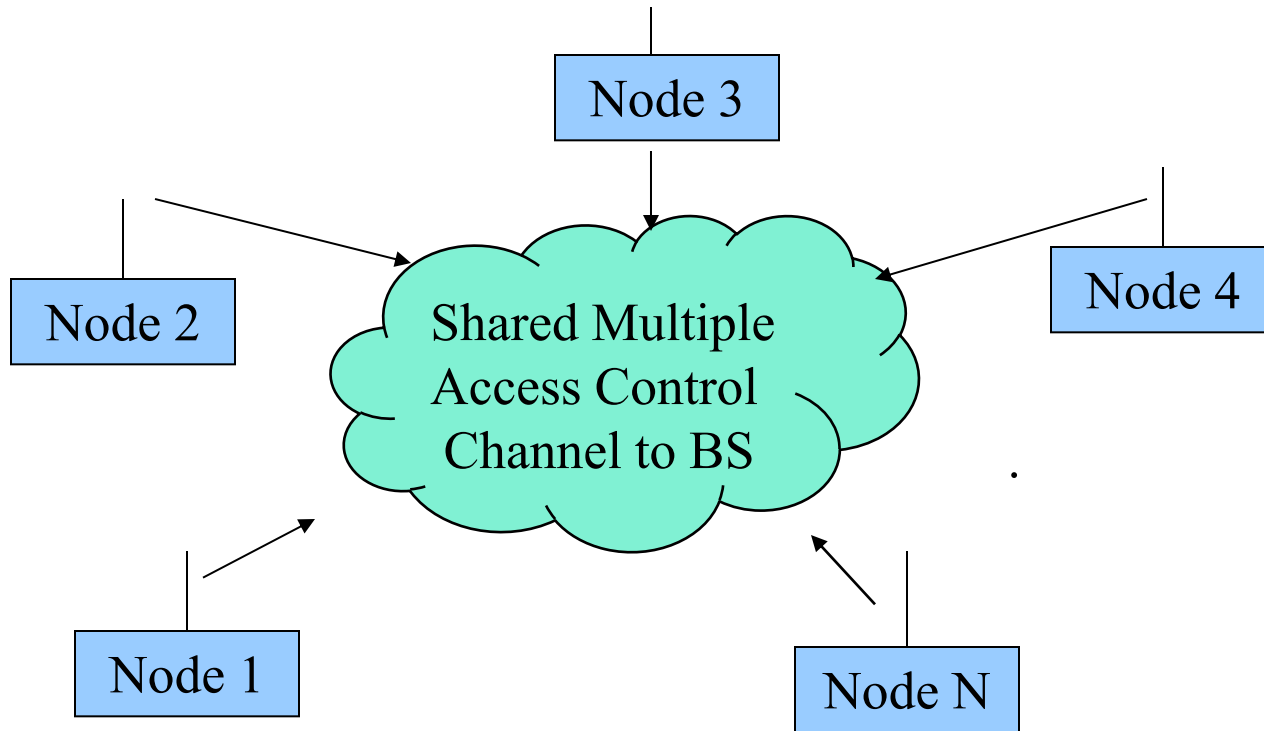


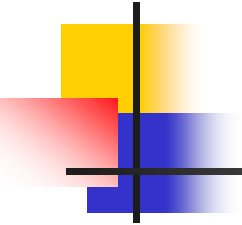
Outline

- Introduction
- Contention Protocols
- ALOHA
- Slotted ALOHA
- CSMA (Carrier Sense Multiple Access)
- CSMA/CD (CSMA with Collision Detection)
- CSMA/CA (CSMA with Collision Avoidance)

Introduction

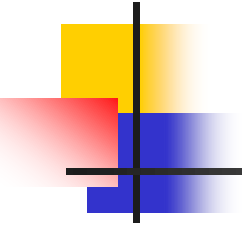
- Multiple access control channels
 - Each node is attached to a transmitter/receiver which communicates via a channel shared by other nodes
 - Transmission from any node is received by other nodes



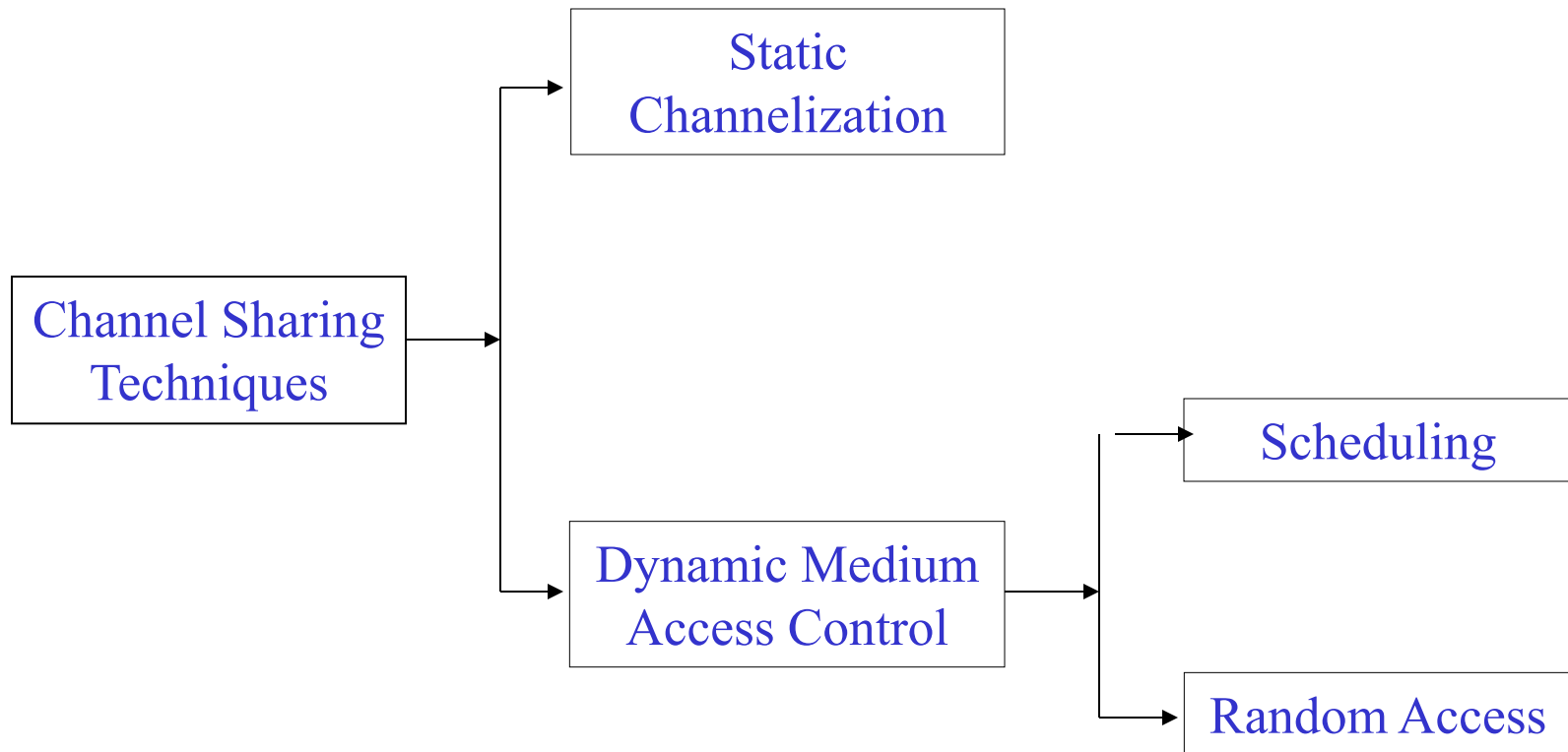


Introduction (Cont'd)

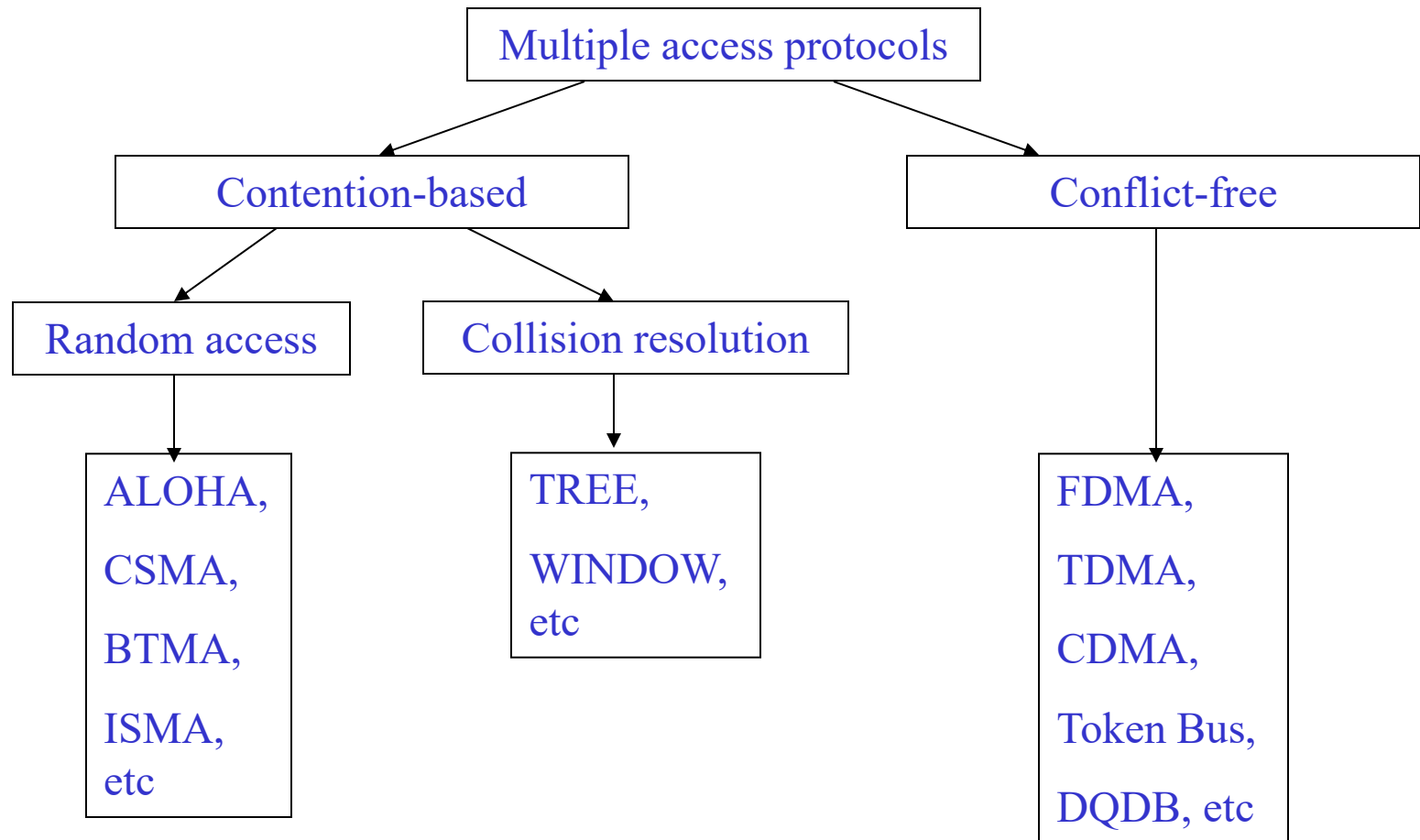
- Multiple access issues
 - If more than one node transmit at a time on the control channel to BS, a collision occurs
 - How to determine which node can transmit to BS?
- Multiple access protocols
 - Solving multiple access issues
 - Different types:
 - Contention protocols resolve a collision after it occurs. These protocols execute a collision resolution protocol after each collision
 - Collision-free protocols (e.g., a bit-map protocol and binary countdown) ensure that a collision can never occur.



Channel Sharing Techniques



Classification of Multiple Access Protocols

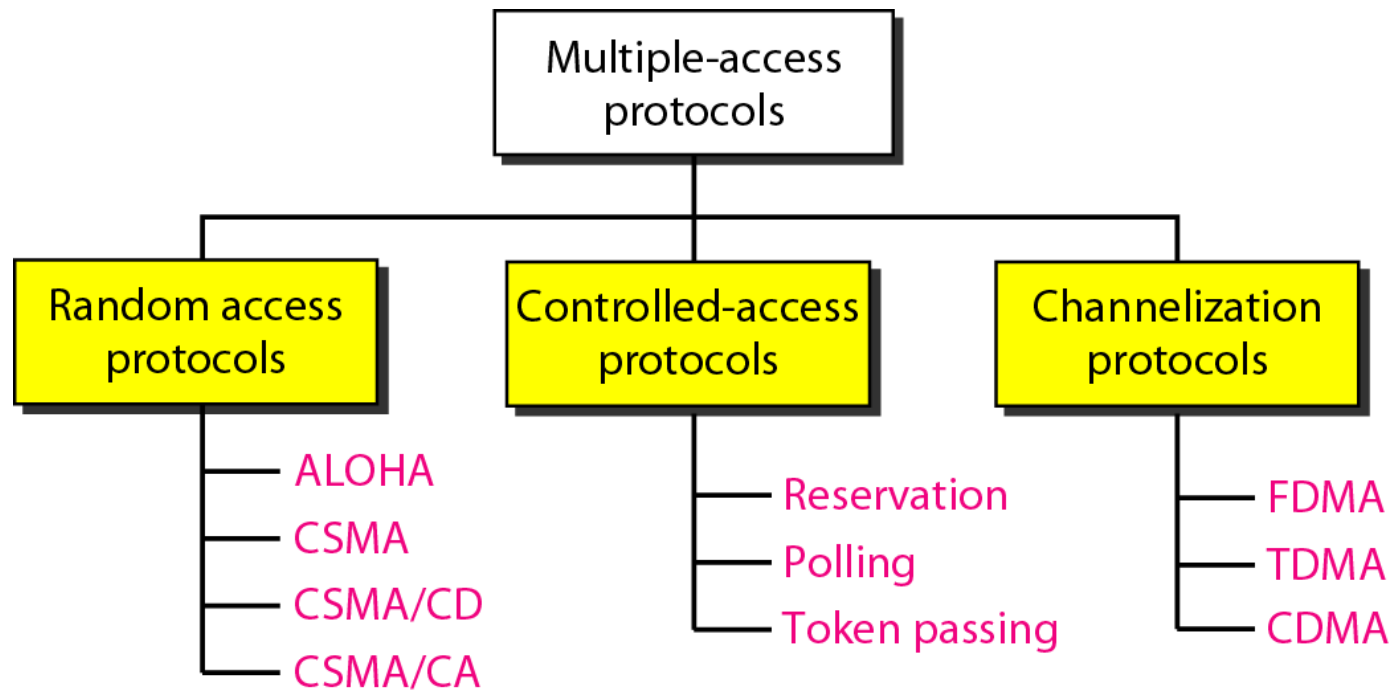


BTMA: Busy Tone Multiple Access

ISMA: Internet Streaming Media Alliance

DQDB: Distributed Queue Dual Bus

Taxonomy of multiple-access protocols discussed



RANDOM ACCESS

In **random access** or **contention** methods, no station is superior to another station and none is assigned the control over another. No station permits, or does not permit, another station to send. At each instance, a station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send.

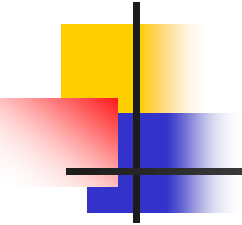
Topics discussed in this section:

ALOHA

Carrier Sense Multiple Access

Carrier Sense Multiple Access with Collision Detection

Carrier Sense Multiple Access with Collision Avoidance



Contention Protocols

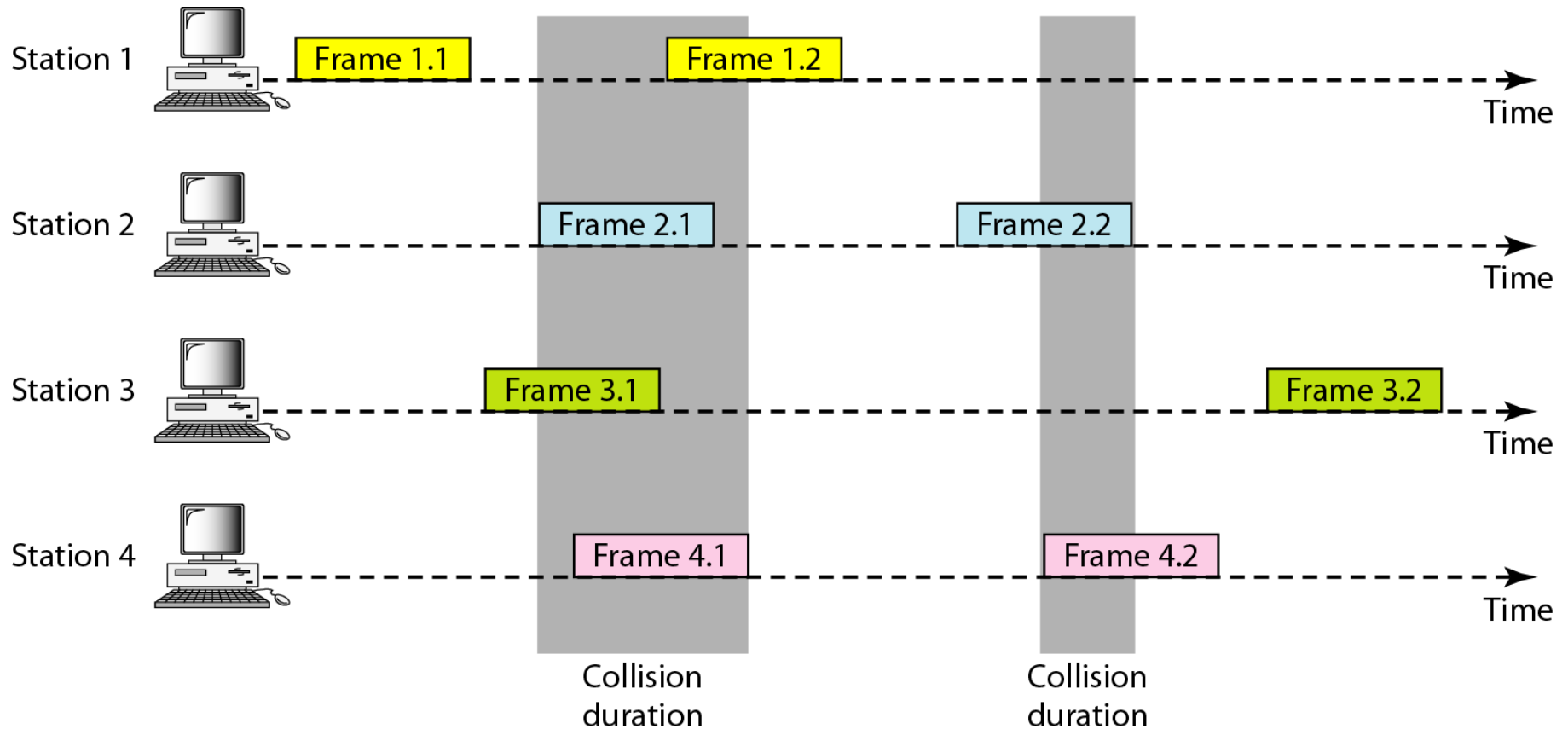
■ ALOHA

- Developed in the 1970s for a packet radio network by Hawaii University.
- Whenever a station has a data, it transmits. Sender finds out whether transmission was successful or experienced a collision by listening to the broadcast from the destination station. Sender retransmits after some random time if there is a collision.

■ Slotted ALOHA

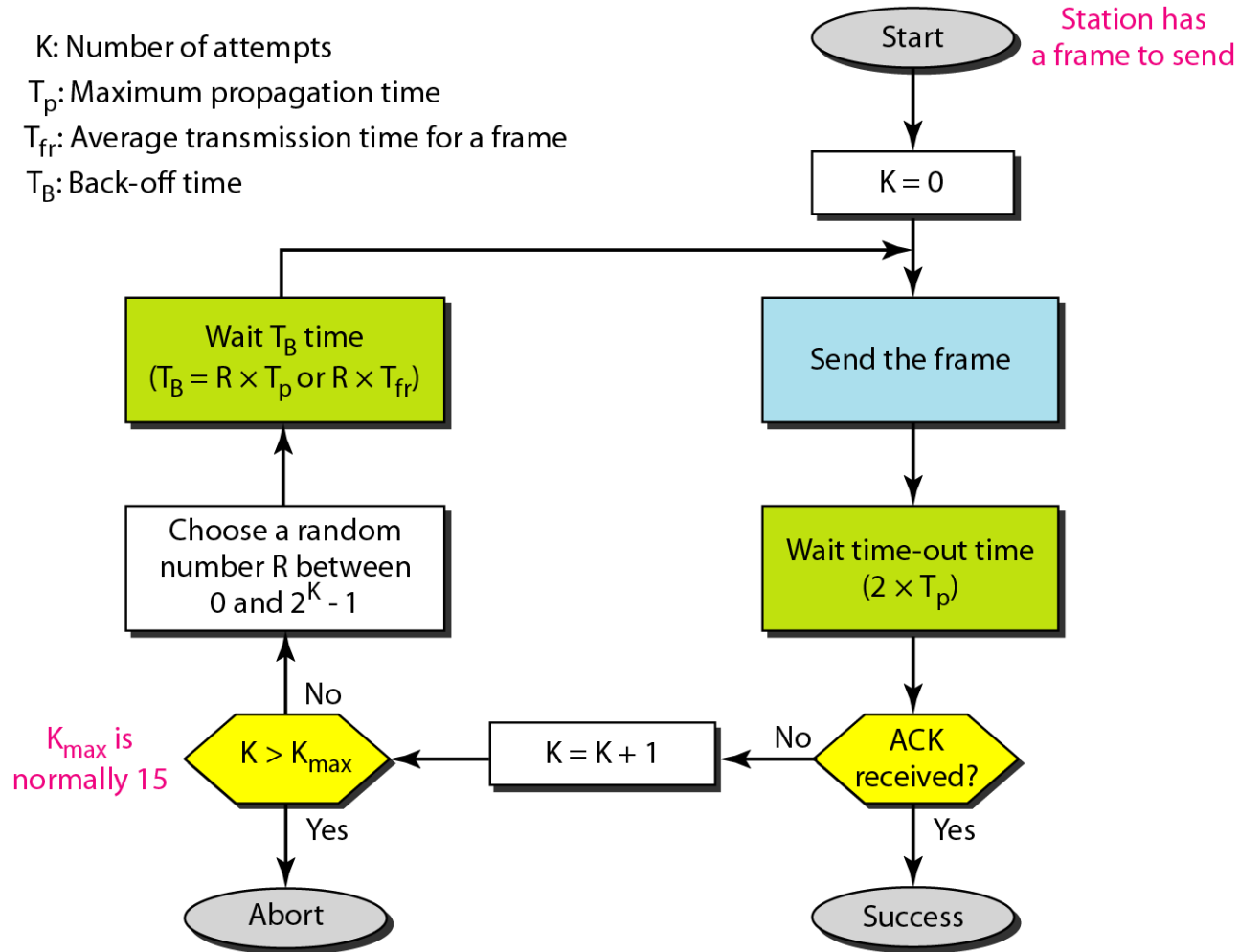
- Improvement: Time is slotted and a packet can only be transmitted at the beginning of one slot. Thus, it can reduce the collision duration.

Frames in a pure ALOHA network



Procedure for pure ALOHA protocol

K: Number of attempts
 T_p : Maximum propagation time
 T_{fr} : Average transmission time for a frame
 T_B : Back-off time



Example 1

The stations on a wireless ALOHA network are a maximum of 600 km apart. If we assume that signals propagate at 3×10^8 m/s, we find

$$T_p = (600 \times 10^3) / (3 \times 10^8) = 2 \text{ ms.}$$

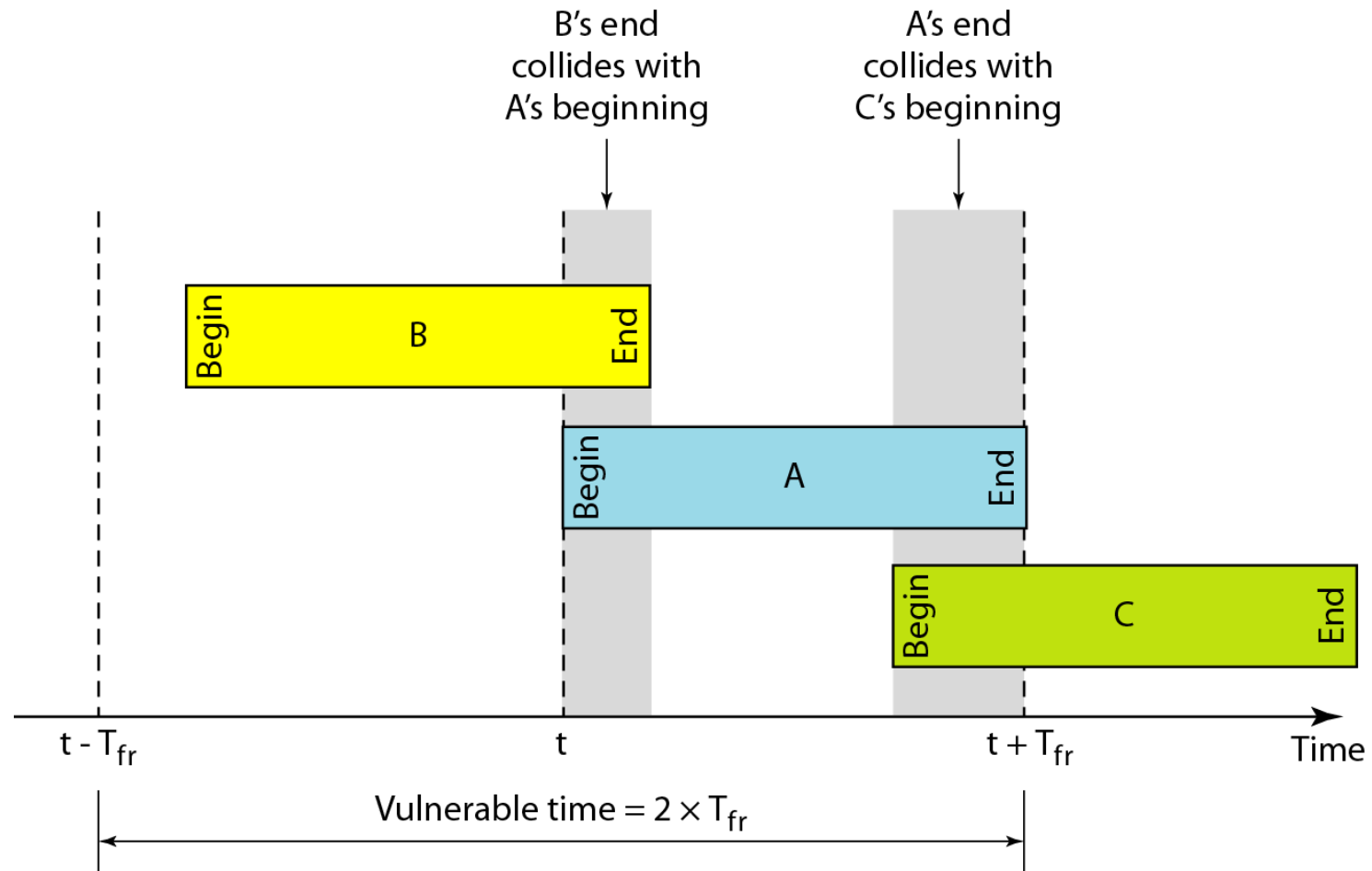
Now we can find the value of T_B for different values of K .

- a. For $K = 1$, the range is $\{0, 1\}$. The station needs to generate a random number with a value of 0 or 1. This means that T_B is either 0 ms (0×2) or 2 ms (1×2), based on the outcome of the random variable.

Example 1 (continued)

- b. For $K = 2$, the range is $\{0, 1, 2, 3\}$. This means that T_B can be 0, 2, 4, or 6 ms, based on the outcome of the random variable.
- c. For $K = 3$, the range is $\{0, 1, 2, 3, 4, 5, 6, 7\}$. This means that T_B can be 0, 2, 4, . . . , 14 ms, based on the outcome of the random variable.
- d. We need to mention that if $K > 10$, it is normally set to 10.

Figure 2 Vulnerable time for pure ALOHA protocol



Example 3

A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the requirement to make this frame collision-free?

Solution

Average frame transmission time T_{fr} is 200 bits/200 kbps or 1 ms. The vulnerable time is $2 \times 1 \text{ ms} = 2 \text{ ms}$. This means no station should send later than 1 ms before this station starts transmission and no station should start sending during the one 1-ms period that this station is sending.



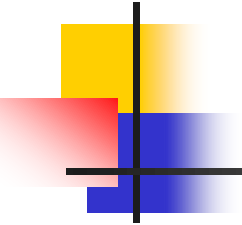
Note

The throughput for pure ALOHA is

$$S = G \times e^{-2G}.$$

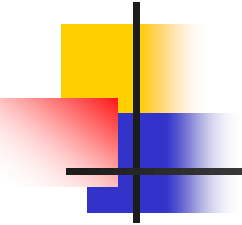
The maximum throughput

$$S_{\max} = 0.184 \text{ when } G = (1/2).$$



Contention Protocols (Cont'd)

- **CSMA** (Carrier Sense Multiple Access)
 - Improvement: Start transmission only if no transmission is ongoing
- **CSMA/CD** (CSMA with Collision Detection)
 - Improvement: Stop ongoing transmission if a collision is detected
- **CSMA/CA** (CSMA with Collision Avoidance)
 - Improvement: Wait a random time and try again when carrier is quiet. If still quiet, then transmit
- **CSMA/CA with ACK**
- **CSMA/CA with RTS/CTS**



Throughput of ALOHA

- The probability that n packets arrive in two packets time is given by

$$P(n) = \frac{(2G)^n e^{-2G}}{n!}$$

where G is traffic load.

- The probability $P(0)$ that a packet is successfully received without collision is calculated by letting $n=0$ in the above equation. We get

$$P(0) = e^{-2G}$$

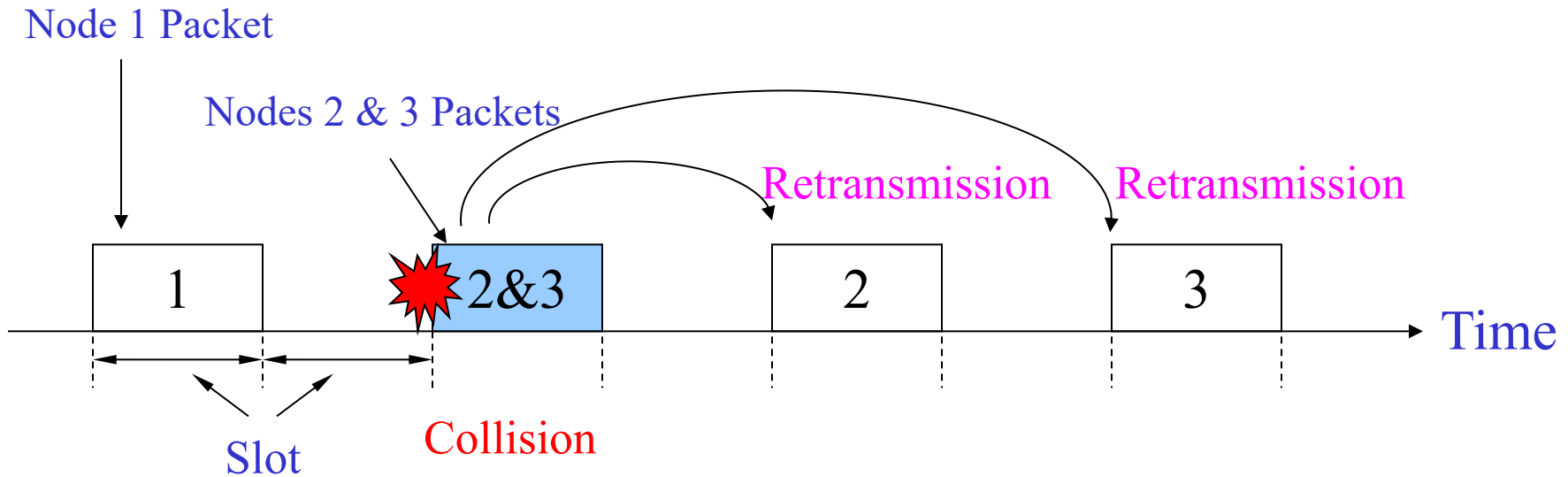
- We can calculate throughput S with a traffic load G as follows:

$$S = G \cdot P(0) = G \cdot e^{-2G}$$

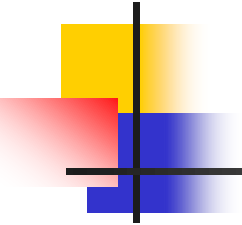
- The Maximum throughput of ALOHA is

$$S_{\max} = \frac{1}{2e} \approx 0.184$$

Slotted ALOHA



Collision mechanism in slotted ALOHA



Throughput of Slotted ALOHA

- The probability of no collision is given by

$$P(0) = e^{-G}$$

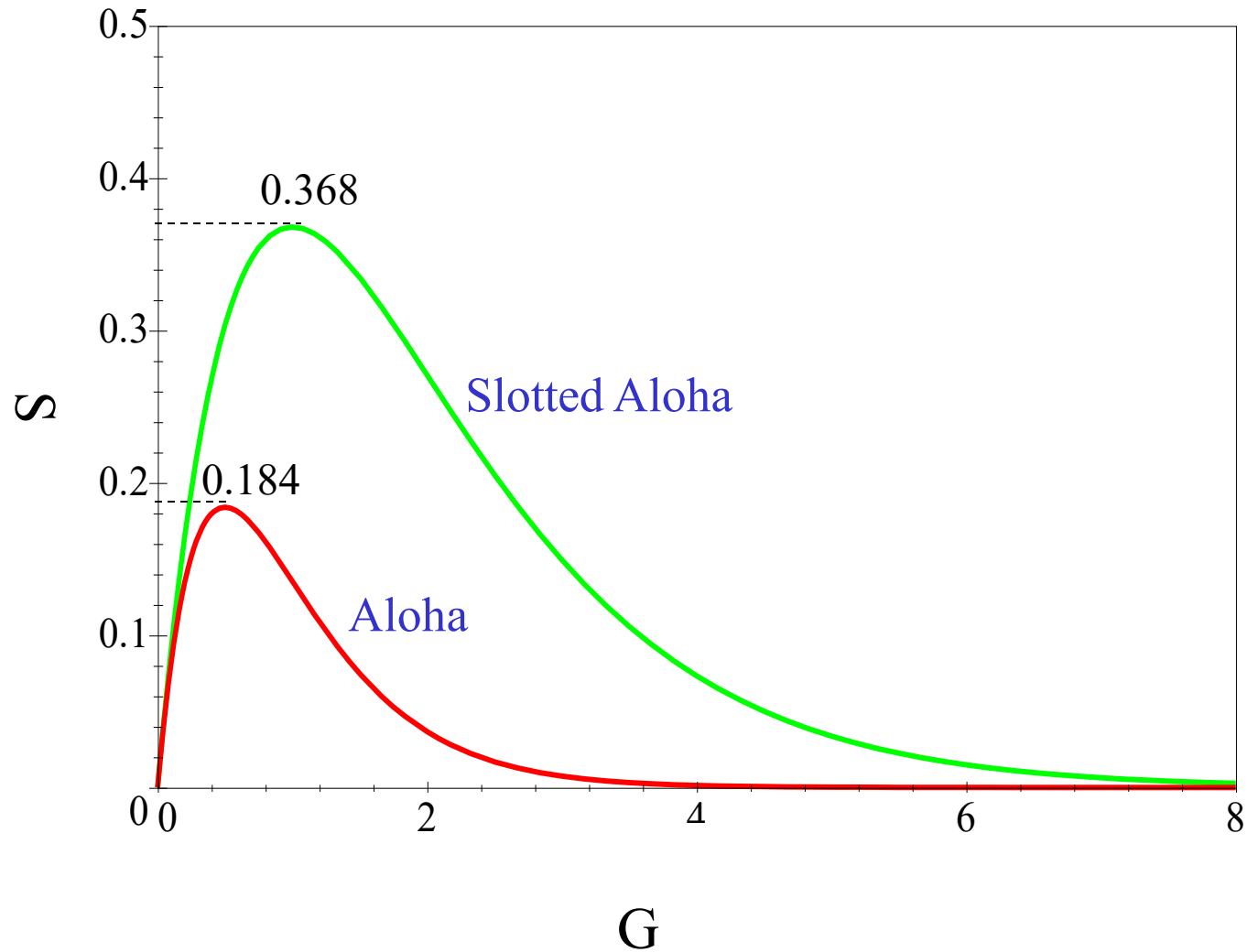
- The throughput S is

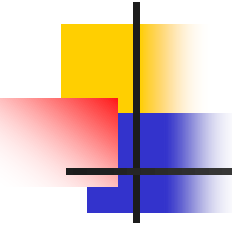
$$S = G \cdot P(0) = G \cdot e^{-G}$$

- The Maximum throughput of slotted ALOHA is

$$S_{\max} = \frac{1}{e} \approx 0.368$$

Throughput

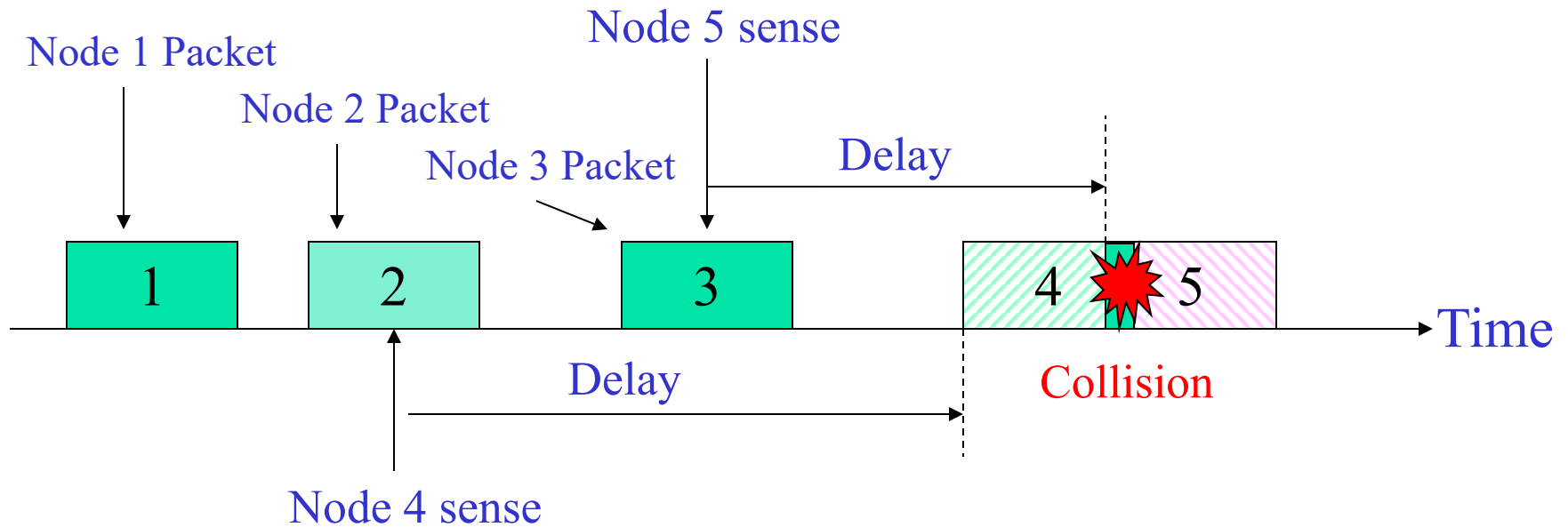




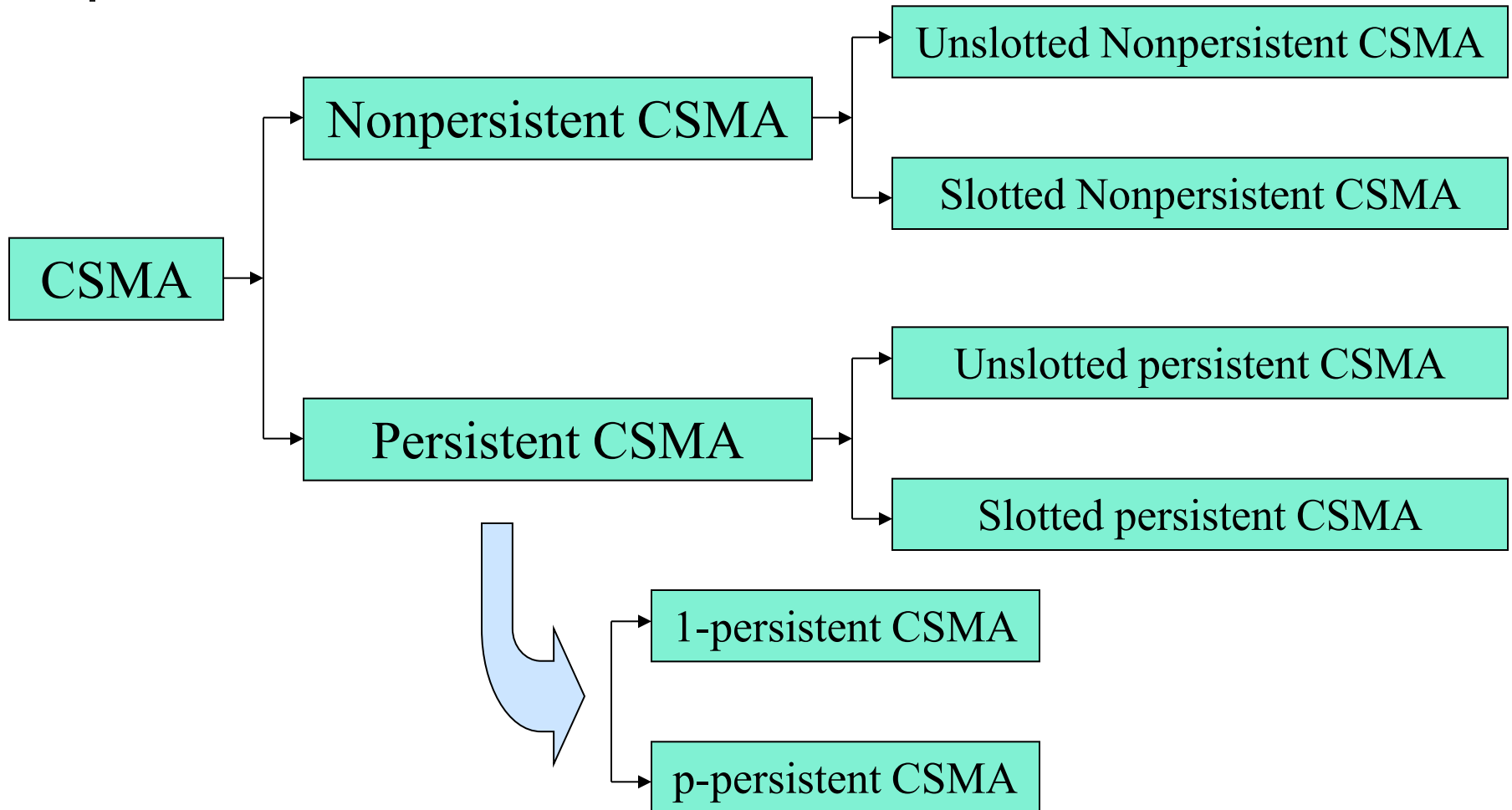
CSMA (Carrier Sense Multiple Access)

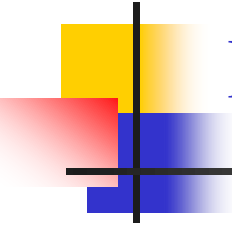
- Max throughput achievable by slotted ALOHA is 0.368.
- CSMA gives improved throughput compared to Aloha protocols.
- Listens to the channel before transmitting a packet (avoid avoidable collisions).

Collision Mechanism in CSMA



Kinds of CSMA





Nonpersistent/x-persistent CSMA Protocols

- Nonpersistent CSMA Protocol:

Step 1: If the medium is idle, transmit immediately

Step 2: If the medium is busy, wait a random amount of time and repeat **Step 1**

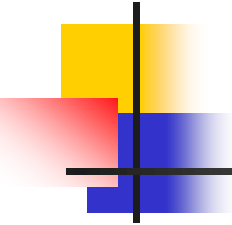
- Random backoff reduces probability of collisions
- Waste idle time if the backoff time is too long

- 1-persistent CSMA Protocol:

Step 1: If the medium is idle, transmit immediately

Step 2: If the medium is busy, continue to listen until medium becomes idle, and then transmit immediately

- There will always be a collision if two nodes want to retransmit (usually you stop transmission attempts after few tries)



Nonpersistent/x-persistent CSMA Protocols

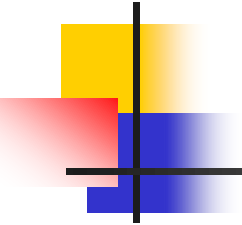
- **p-persistent CSMA Protocol:**

Step 1: If the medium is idle, transmit with probability p , and delay for worst case propagation delay for one packet with probability $(1-p)$

Step 2: If the medium is busy, continue to listen until medium becomes idle, then go to **Step 1**

Step 3: If transmission is delayed by one time slot, continue with **Step 1**

- A good tradeoff between nonpersistent and 1-persistent CSMA

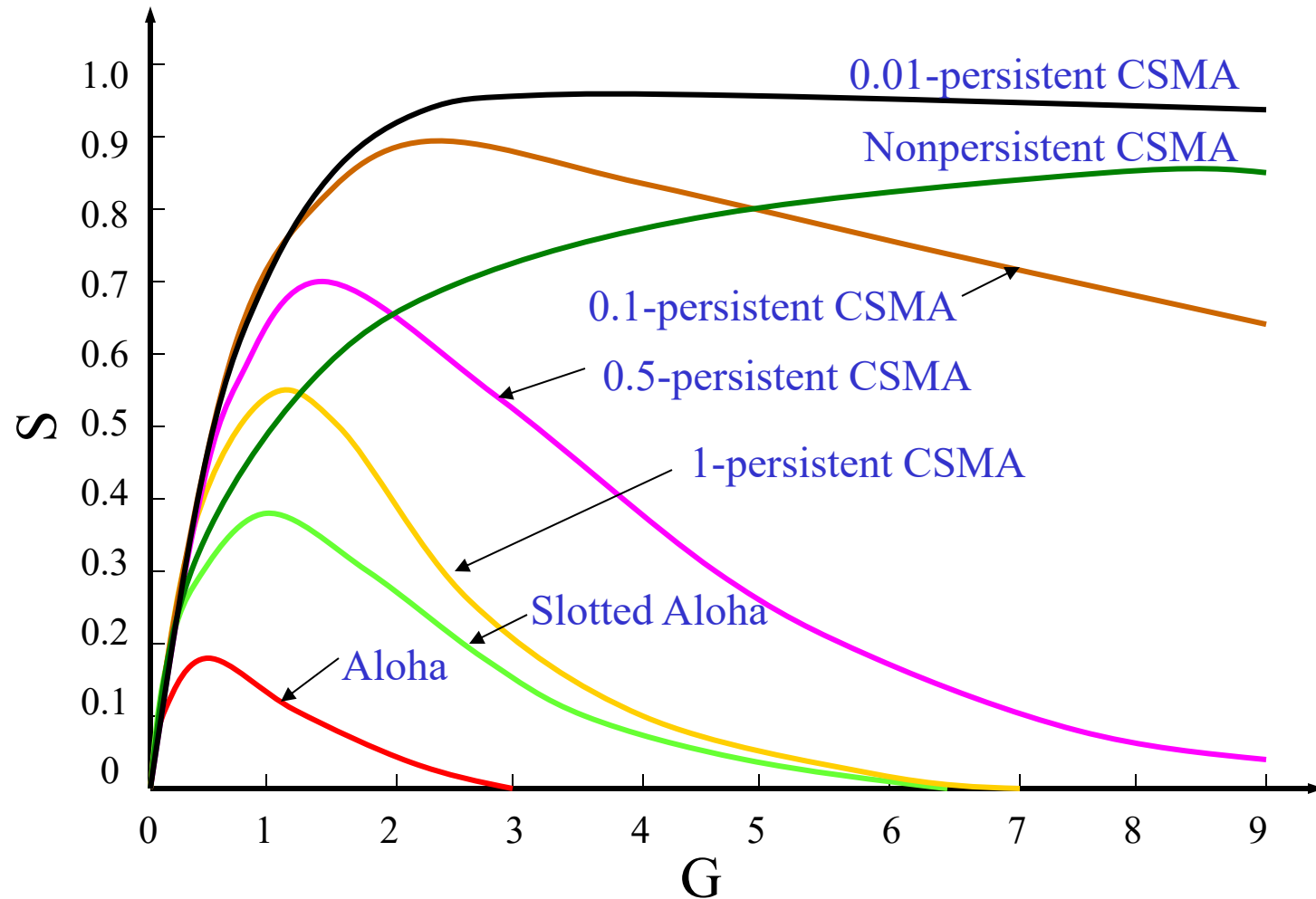


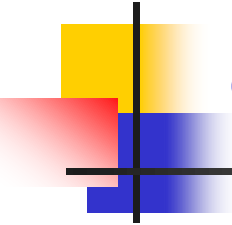
How to Select Probability p ?

- Assume that N nodes have a packet to send and the medium is busy
- *Then*, Np is the expected number of nodes that will attempt to transmit once the medium becomes idle
- If $Np > 1$, then a collision is expected to occur

Therefore, network must make sure that $Np < 1$ to avoid collision, where N is the maximum number of nodes that can be active at a time

Throughput

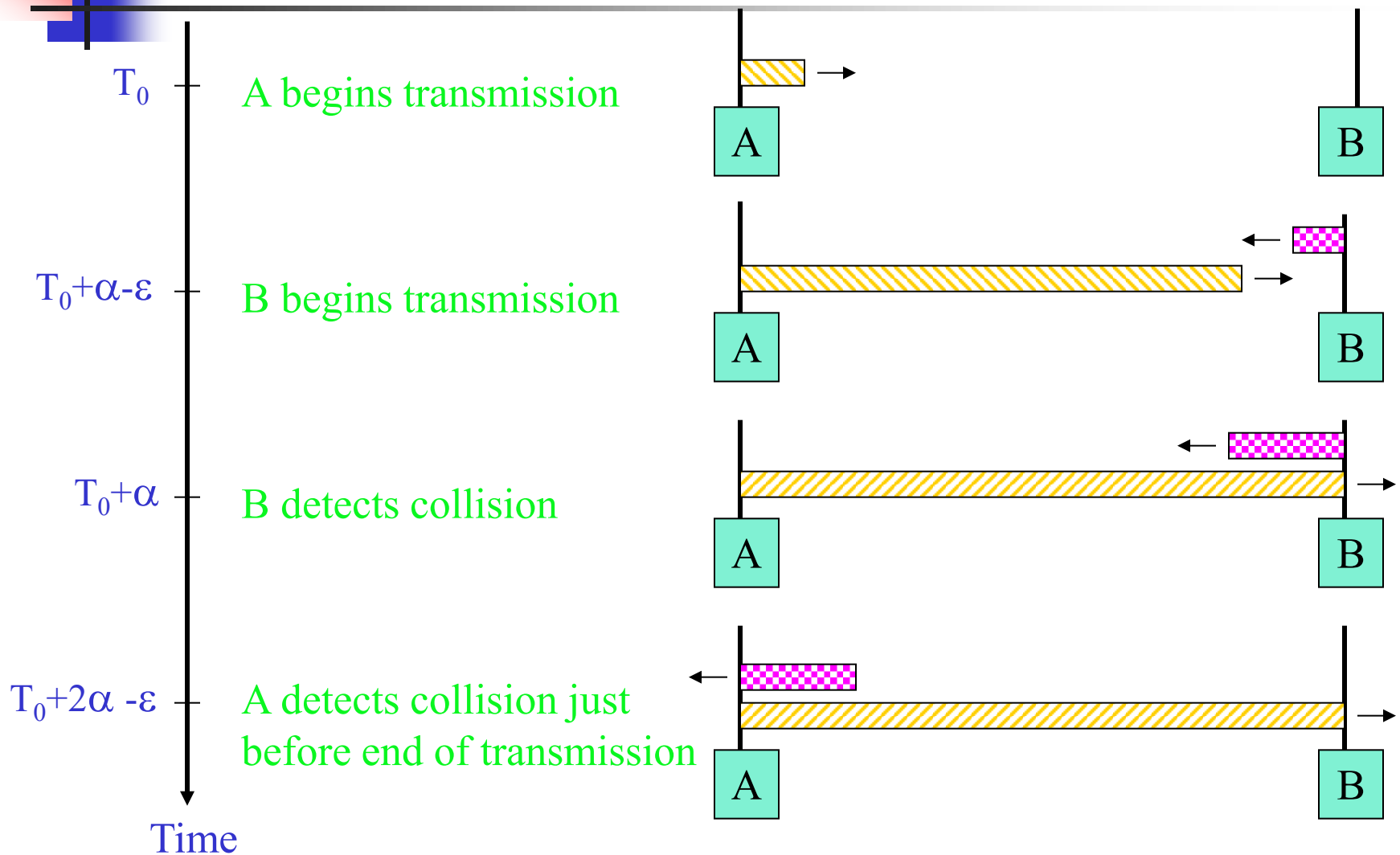




CSMA/CD (CSMA with Collision Detection)

- In CSMA, if 2 terminals begin sending packet at the same time, each will transmit its complete packet (although collision is taking place).
- Wasting medium for an entire packet time.
- CSMA/CD
 - Step 1: If the medium is idle, transmit
 - Step 2: If the medium is busy, continue to listen until the channel is idle then transmit
 - Step 3: If a collision is detected during transmission, cease transmitting
 - Step 4: Wait a random amount of time and repeats the same algorithm

CSMA/CD (Cont'd)



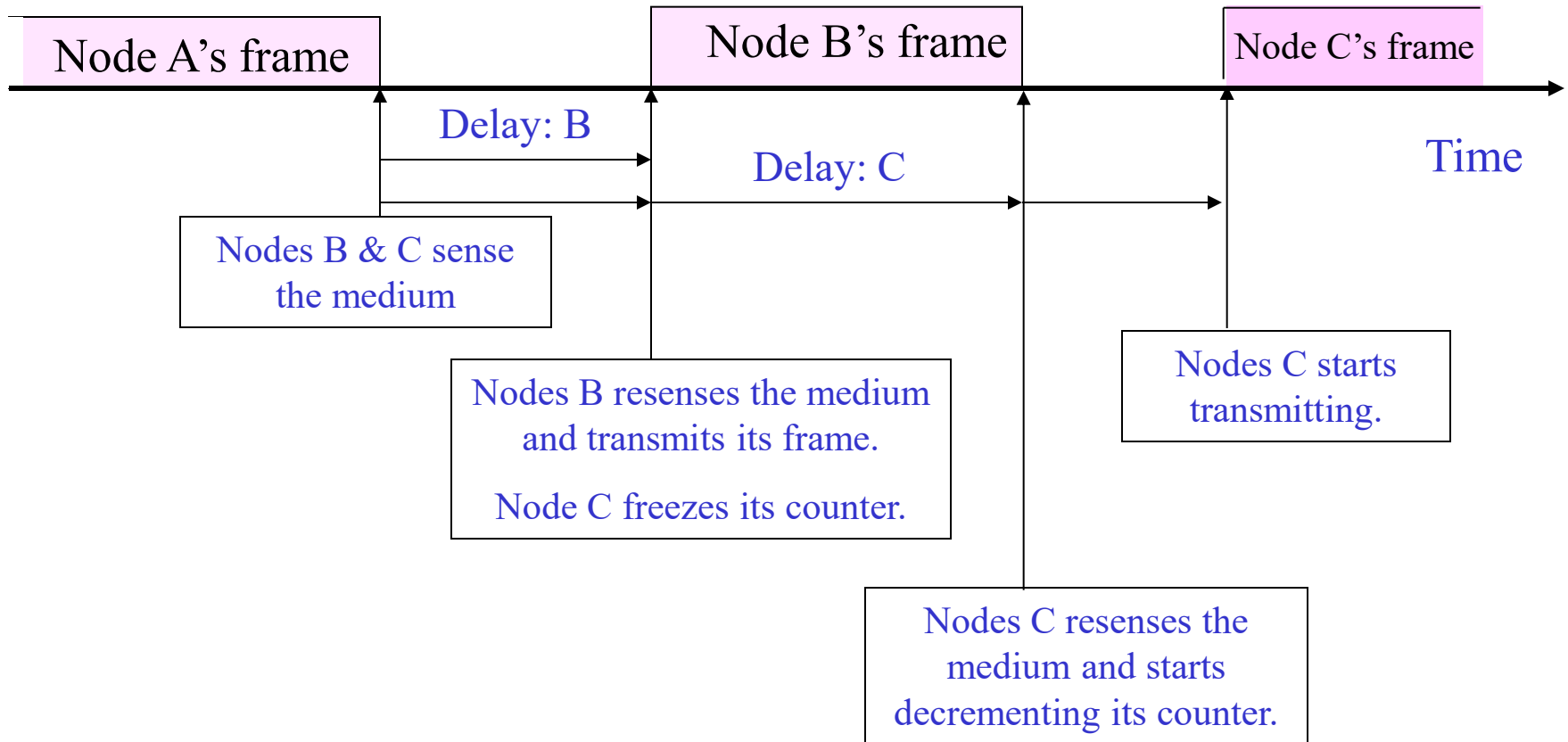
(α is the propagation time)



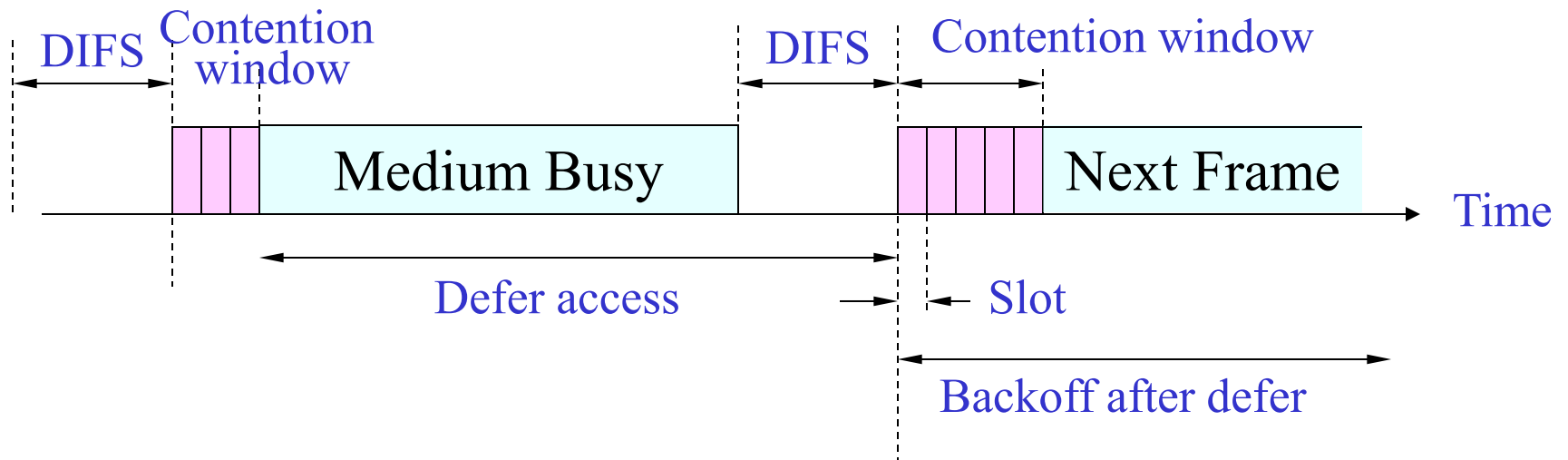
CSMA/CA (CSMA with collision Avoidance)

- All terminals listen to the same medium as CSMA/CD.
- Terminal ready to transmit senses the medium.
- If medium is busy it waits until the end of current transmission.
- It again waits for an additional predetermined time period DIFS (Distributed inter frame Space).
- Then picks up a random number of slots (the initial value of backoff counter) within a contention window to wait before transmitting its frame.
- If there are transmissions by other terminals during this time period (backoff time), the terminal freezes its counter.
- It resumes count down after other terminals finish transmission + DIFS. The terminal can start its transmission when the counter reaches to zero.

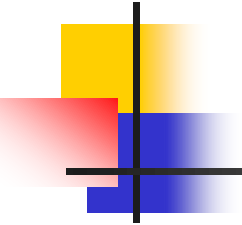
CSMA/CA (Cont'd)



CSMA/CA Explained



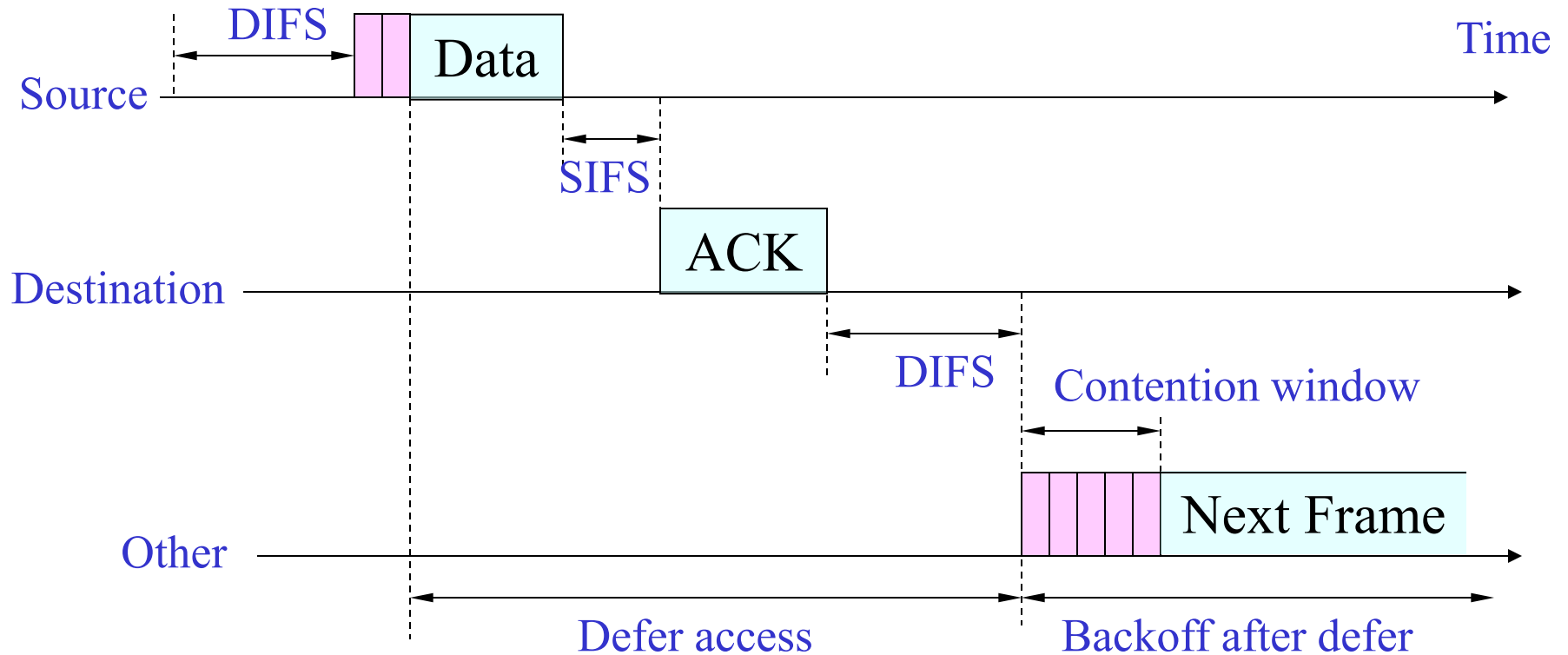
DIFS – Distributed Inter Frame Spacing



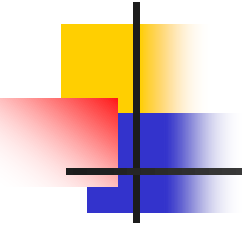
CSMA/CA with ACK

- Immediate Acknowledgements from receiver upon reception of data frame without any need for sensing the medium.
- ACK frame transmitted after time interval SIFS (*Short Inter-Frame Space*) ($SIFS < DIFS$)
- Receiver transmits ACK without sensing the medium.
- If ACK is lost, retransmission done.

CSMA/CA/ACK



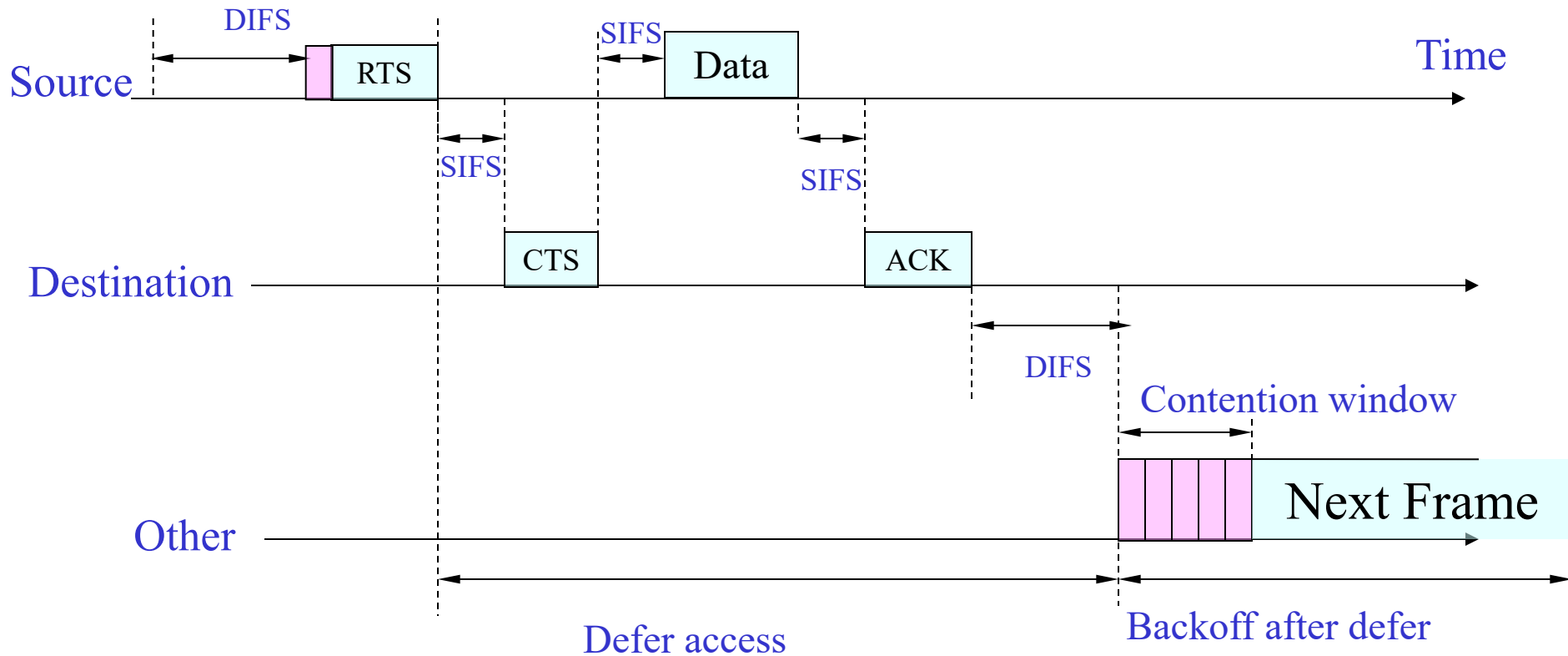
SIFS – Short Inter Frame Spacing



CSMA/CA with RTS/CTS

- Transmitter sends an RTS (request to send) after medium has been idle for time interval more than DIFS.
- Receiver responds with CTS (clear to send) after medium has been idle for SIFS.
- Then Data is exchanged.
- RTS/CTS is used for reserving channel for data transmission so that the collision can only occur in control message.

CSMA/CA with RTS/CTS (Cont'd)



RTS/CTS

